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spherical: A Comprehensive Database and Automated Pipeline for VLT/SPHERE High-Contrast Imaging

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Software

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Summary

The Spectro-Polarimetric High-contrast Exoplanet REsearch instrument (SPHERE, Beuzit et al., 2019) at the Very Large Telescope (VLT) is one of the main facilities for coronagraphic imaging of exoplanets and circumstellar disks in the optical and near-infrared. Over the last decade, SPHERE has contributed to hundreds of publications and major legacy surveys (e.g., the SpHere INfrared survey for Exoplanets, SHINE, Chauvin et al., 2017; Chomez et al., 2025).

`spherical` combines a curated, searchable database of the complete SPHERE observation history with automated reduction pipelines for two of its subsystems. Users can filter observations by target star properties, observing mode, or observing conditions, download the selected raw data and calibrations from the European Southern Observatory (ESO) archive, and reduce them end to end. Figure 1 shows the coverage of the database and one observation taken from the archive to a detection and a spectrum. Integral Field Spectrograph (IFS, Claudi et al., 2008) data are extracted into spectral cubes by a pipeline adapted from the Coronagraphic High Angular Resolution Imaging Spectrograph (CHARIS, Brandt et al., 2017; Samland et al., 2022). Data from the Infra-Red Dual-band Imager and Spectrograph (IRDIS, Dohlen et al., 2008) taken in dual-band (DBI) or broadband imaging are calibrated and pre-processed by a second pipeline that follows the same steps. Both feed post-processing with TRAP (Temporal Reference Analysis of Planets, Samland et al., 2021), a detection algorithm that identifies companions by modeling the temporal systematics of the starlight instead of subtracting a reference image, and recovers their astrometry and contrast spectra including their respective covariances.

The v3.2.0 database release (Samland, 2026) covers observations taken between 2014 May and 2026 October. It lists 4384 IFS sequences and 5585 IRDIS sequences, most of the latter in dual-band or broadband imaging. A further 1155 IRDIS sequences use dual-beam polarimetric imaging (DPI, de Boer et al., 2020; van Holstein et al., 2020a), and 355 sequences from both subsystems use sparse aperture masking (SAM, Cheetham et al., 2016). Another 393 sequences of solar-system bodies, galaxies, and other non-stellar targets are kept in separate tables. DPI and SAM data are covered for discovery and download only. Their reduction is left to community tools. SPHERE is undergoing the *SPHERE+* upgrade (Boccaletti et al., 2022, 2020), including a second-stage adaptive optics system (SAXO+, Stadler et al., 2022), which extends its scientific lifetime and makes it a pathfinder for the Planetary Camera and Spectrograph (PCS, Kasper et al., 2021) of the Extremely Large Telescope (ELT).

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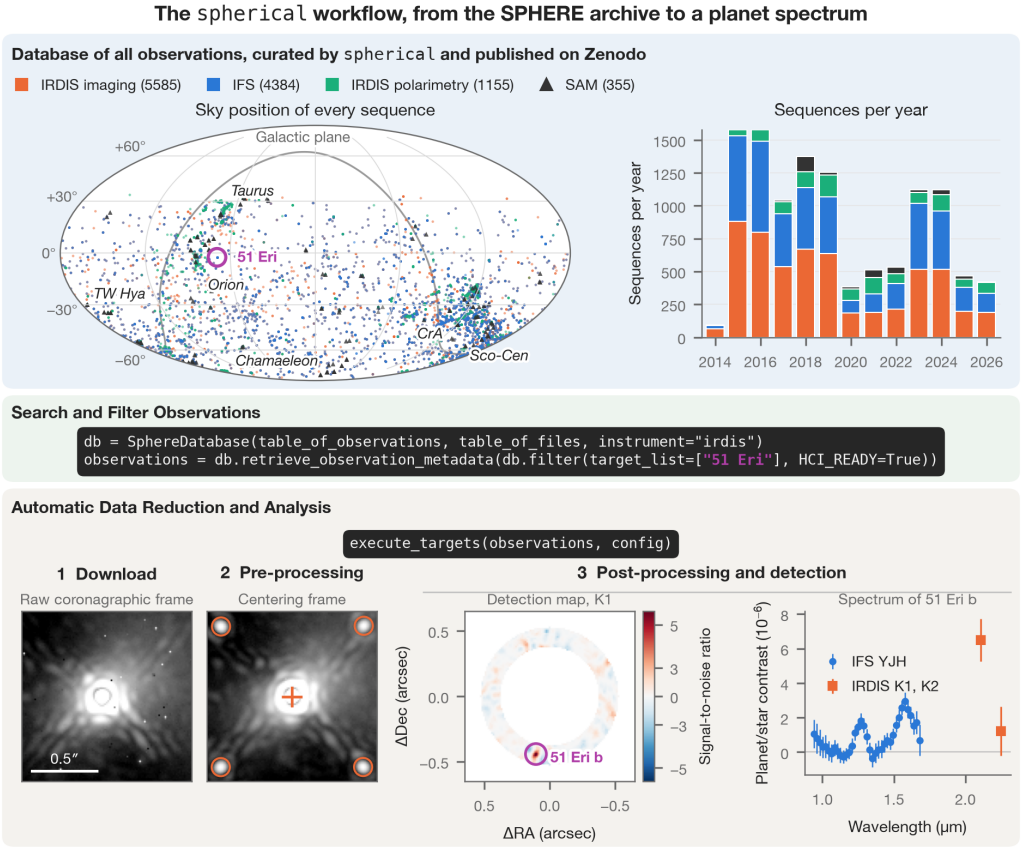


Figure 1: The upper box shows the v3.2.0 database, which covers 2014 May to 2026 October. The sky map uses equatorial coordinates (Mollweide projection, right ascension 0h at the centre and increasing to the left) and labels young star-forming regions and OB associations identified through MOCAdb. Young moving groups such as the β Pictoris moving group are spread over the whole sky and are not labelled. The years 2014 and 2026 are partial. The lower box shows the simultaneous IRDIS (K12) and IFS (YJH) observation of 51 Eri on the night of 2015 September 24. Its panels show one raw 16 s IRDIS frame in K1 in detector orientation, a centering frame whose satellite spots (circles) locate the star behind the coronagraph (cross), the TRAP detection map in K1 (north up, east left, limited to the search annulus), and the contrast spectrum of 51 Eri b. All images use non-linear colour scales.

Statement of Need

The ESO VLT/SPHERE archive is the world's largest collection of high-contrast imaging data for detecting exoplanets, substellar companions, and circumstellar disks. Using it end to end remains difficult. The observation history is not available in a form that can be queried by scientific criteria, only by header keywords such as target name or coordinates. Raw data and their calibrations must be queried and retrieved by hand. These barriers are most challenging for researchers assembling large homogeneous samples for population studies or for uniform extraction of exoplanet spectra, and for survey teams needing rapid follow-up. `spherical` is written for observers working on direct imaging of exoplanets and disks, SPHERE survey teams, and archival researchers.

State of the Field

Several existing tools address isolated components of the SPHERE data workflow. The **High Contrast Data Center (DC)** (Delorme et al., 2017) reduces datasets with the ESO pipeline

recipes behind a Java-based interface, but offers limited capabilities for programmatic interaction or custom batch processing. **vlt-sphere** (Vigan, 2020) drives the same recipes through `esorex` for user-provided raw data, and therefore requires a local ESO pipeline installation. It lacks automated archival download and integrated post-processing. For polarimetry, **IRDAP** (van Holstein et al., 2020b) reduces the data automatically, but the user has to find and retrieve the raw data.

None of them gives an overview of what SPHERE has observed over its lifetime. `spherical` provides that overview and connects it to one workflow, from the selection of observations through download and reduction to the detection of companions and the extraction of their spectra with TRAP. Its retrieval and pre-processing stages can also feed other post-processing packages such as VIP (Christiaens et al., 2023; Gomez Gonzalez et al., 2017), pyKLIP (Wang et al., 2015), and IRDAP.

Software Design

`spherical` is built as a layer over the ESO raw archive that keeps metadata curation separate from data reduction. Science headers ingested from the ESO archive are cross-matched on telescope pointing against SIMBAD (Wenger et al., 2000) and Gaia DR3 (Gaia Collaboration et al., 2023). Because archival metadata is noisy, target ambiguities are resolved by proximity, brightness, and object type, with stellar proper motion propagated to the epoch of observation. Each sequence is assigned to a single host star, and other catalogued stars in the field are listed alongside it.

The cross-match produces a structured, searchable observation table, available pre-computed on Zenodo (DOI: 10.5281/zenodo.15147730). It aggregates observing mode, total exposure time, parallactic angle coverage, and atmospheric conditions. Accompanying target tables carry Gaia DR3 astrophysical parameters and, via a cross-match against MOCAdb (Gagné et al., 2026), young-association membership, BANYAN Σ membership probabilities, and adopted ages. With these columns, samples can be selected by stellar age and association membership as well as by pointing, conditions, and stellar parameters. Choosing a local, curated table over live archive queries makes complex filtering of the entire instrument history fast. The tables can be built from scratch, but the recommended entry point is the published set, which a single command extends with data taken since the last release. Every build records its provenance, both in a separate file and in FITS header keywords. The record includes the `spherical` version, the ESO query date and coverage range, the Gaia data release, the status of each enrichment source, and the build parameters.

`spherical` wraps the existing reduction algorithms. It provides a Python interface to the IFS cube extraction of Samland et al. (2022) and to TRAP and combines both in one object-oriented framework. Code written for single datasets thereby runs over hundreds of them without changes to the algorithms or their published behavior. This keeps results comparable with earlier work and limits the code that has to be maintained. The reduction does not depend on ESO's pipeline software. Calibration, extraction, and pre-processing are implemented in Python. The full chain installs with `pip` without `esorex` or the ESO pipeline kits, and every step can be inspected and modified in one language.

The IRDIS imaging pipeline, introduced in v3.0.0, follows the same steps as the IFS pipeline. It builds master calibrations for each observation and pre-processes the data into calibrated cubes with analytic inverse-variance maps. The star center is measured on the satellite-spot (waffle) frames, and for dithered sequences the offsets are propagated through the dithering header keywords. Photometric calibration uses the unocculted point spread function (PSF) without saturated flux frames, and TRAP post-processing matches spectral templates across the two channels. Both subsystems share one configuration model and by default resume an interrupted campaign where it stopped. The pipelines are validated against a benchmark dataset, the simultaneous IRDIS and IFS observation of 51 Eridani b from 2015 September

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(Samland et al., 2017). The astrometry and photometry of the planet recovered from both data sets agree within the uncertainties with the published values (Maire et al., 2019; Samland et al., 2017), and regression tests pin both results to frozen baselines. The documentation presents both reductions as end-to-end tutorials.

`spherical` builds only on community Python packages. It uses `Astropy` (Astropy Collaboration et al., 2013, 2018, 2022), `astroquery` (Ginsburg et al., 2019) for access to the ESO archive and to catalogs, `NumPy` (Harris et al., 2020) for numerical operations, and pandas (The pandas development team, 2020) for tables. The IFS reduction uses the adapted CHARIS pipeline (Brandt et al., 2017; Samland et al., 2022), with calibration routines derived in part from Vigan (2020) (see the `vlt-sphere` repository for individual contributors). Stellar effective temperatures for TRAP's spectral templates come from Gaia DR3 where available and otherwise from a tabulation of Pecaut & Mamajek (2013). Post-processing employs TRAP (Samland et al., 2021).

Research Impact Statement

Before this publication, `spherical` was used to extract spectra of exoplanet atmospheres from SPHERE IFS data in several peer-reviewed studies (Franson et al., 2023; Hammond et al., 2025; Stolker et al., 2025). At the time these works were published, the code was not yet citable via an existing DOI. Since the database of existing observations was made available on Zenodo, it has been downloaded more than 900 times. New exoplanet candidates identified with `spherical` have been accepted for follow-up observations in ESO period P117 (programme 117.2A06.001). The programme received 28.5 hours, more than 60% of the time allocated to SPHERE in P117.

Reducing the effort needed to go from raw archival data to science-ready products makes it practical to assemble the homogeneous samples that exoplanet occurrence-rate and population studies require, and to reprocess the full twelve-year archive with one algorithm in search of companions missed by earlier, less sensitive techniques. Because database releases are versioned and record their provenance, an analysis can name the database state it used, and others can retrieve that state and repeat the selection.

Future Work

Database coverage and retrieval will be extended to the remaining observing modes, the Zurich Imaging Polarimeter (ZIMPOL, Schmid et al., 2018) and IRDIS long-slit spectroscopy (Vigan et al., 2008), with the respective community reduction pipelines integrated into the same script-driven framework. The package will also follow the SPHERE+ upgrade, with calibration and post-processing adapted to the improved instrument and to SAXO+.

AI Usage Disclosure

Generative AI was used in the development of `spherical` and in the preparation of this manuscript. The system used was Claude Code (Anthropic), with Claude Sonnet and Claude Opus models of the 4.x and 5 model families, over the period from mid-2025 to the v3.2.0 release in October 2026.

Within the software, AI assistance went mainly into the infrastructure around the scientific algorithms. This covers pipeline orchestration and step sequencing, structured logging and crash reports, the typed configuration dataclasses, the console entry points, and the workflow that builds and updates the database and records its provenance. AI was also used for the test suite and for docstrings, README, and changelog text. In preparing this manuscript,

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generative AI was used for language editing and to improve conciseness, not to draft its scientific content.

AI was not used for the scientific methodology. The spectral extraction (Samland et al., 2022) and post-processing (Samland et al., 2021) algorithms are pre-existing published work and are wrapped unchanged. The calibration methodology, the target-resolution logic that disambiguates archival pointings, the database schema, the decision to wrap the existing algorithms, and the validation strategy are the author's own.

The author reviewed, edited, and validated all AI-assisted output, and made the primary architectural and scientific design decisions. Correctness was tested on scientific results, through the 51 Eridani b benchmark and the regression tests described above, in addition to review of the code.

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